

# Autosomal Dominant Osteopetrosis Type I (ADO-I): A Comprehensive Disease Characterization

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**Disease:** Autosomal Dominant Osteopetrosis Type I (ADOI) **Primary identifiers:** OMIM #607634 · Orphanet ORPHA:53 (Osteopetrosis, autosomal dominant) · MeSH D010022 (Osteopetrosis) · Proposed MONDO: high-bone-mass/LRP5 osteopetrosis spectrum **Causal gene:** *LRP5* (HGNC:6697), 11q13 **Category:** Mendelian (autosomal dominant)

*Evidence base: 44 literature records reviewed; 10 confirmed findings. Primary citations given as PMIDs. Evidence source types: human clinical, in vitro, mouse model, structural/computational.*

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## Summary

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Autosomal Dominant Osteopetrosis Type I (ADO-I) is a rare, mild, autosomal-dominant sclerosing bone dysplasia that has been mechanistically reclassified over the past two decades. Originally grouped with the "osteoclast-failure" osteopetroses, ADO-I is now understood to be a **high-bone-mass (HBM) disorder caused by heterozygous gain-of-function missense mutations in *LRP5***, the Wnt co-receptor gene on chromosome 11q13. This distinction is not merely nosological: it explains nearly every clinical peculiarity of the disease, above all the fact that **ADO-I is the only osteopetrosis that does *not* carry an increased fracture risk**.

The unifying mechanism is a loss of receptor braking. Normally the Wnt inhibitors **sclerostin (SOST)** and **Dickkopf-1 (DKK1)** dock onto the first  $\beta$ -propeller (E1) domain of *LRP5* and shut down Wnt/ $\beta$ -catenin signaling in osteoblasts. ADO-I mutations (G171V, D111Y, G171R, A214T/V, A242T, T253I) cluster precisely in this E1 docking module and prevent inhibitor binding. The receptor therefore signals constitutively, osteoblasts over-build matrix, and bone accrues progressively — cortical bone in particular — with the skull vault most severely affected. Osteoclasts are intrinsically normal, confirming that ADO-I is an osteoblast/Wnt-driven condition rather than a resorption defect. This biology is the exact molecular mirror image of the anabolic osteoporosis drugs (anti-sclerostin romosozumab, anti-DKK1 agents) developed to *raise* bone mass in fragile patients.

Clinically, ADO-I is frequently asymptomatic ( $\approx 39\%$  of gene carriers) and often discovered incidentally on radiographs. When morbidity does occur, it arises predominantly from **cranial and skull-base hyperostosis crowding the neural foramina and posterior fossa**, producing cranial-nerve compression syndromes — hearing loss, trigeminal neuralgia, facial palsy, and visual compromise — rather than from the fractures that dominate other osteopetroses. Inheritance is autosomal dominant with **incomplete penetrance and highly variable, age-progressive expressivity**. Diagnosis is radiologic (skull-vault and cortical osteosclerosis on X-ray/CT plus DXA quantification) confirmed by *LRP5* sequencing, and management is supportive and surgical; hematopoietic stem-cell transplantation, the treatment of choice for the osteoclast-defect forms, is mechanistically inapplicable here.

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## Key Findings

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### Finding 1 — ADO-I is an *LRP5* gain-of-function high-bone-mass disorder (11q13) [human clinical]

Linkage analysis in two Danish ADO-I families mapped the disease locus to chromosome **11q12–13** (summed maximum LOD +6.54 with marker D11S1889; candidate region 6.6 cM between D11S1765 and D11S4113), the interval containing *LRP5* ([PMID: 12054167](#)). Van Wesenbeeck and colleagues then identified **six novel heterozygous missense mutations** — D111Y, G171R, A214T, A214V, A242T, and T253I — in the N-terminal region of *LRP5* across a spectrum of increased-bone-density conditions including osteopetrosis type I. All cluster in the same region as the previously reported G171V HBM mutation; the mutations are "*located in the aminoterminal part of the gene*" before the first EGF-like/ $\beta$ -propeller domain ([PMID: 12579474](#)). The recognition of these gain-of-function alleles **prompted a formal reclassification**: "*The identification of *LRP5* gain-of-function mutations in autosomal dominant osteopetrosis type I prompted a revision of the classification scheme, and this form is now being included among the high-bone-mass diseases*" ([PMID: 20855225](#)).

This is the central reframing of the disease: ADO-I is not a defect of bone *removal* but of excessive bone *formation* driven by a Wnt co-receptor.

## Finding 2 — Mechanism: mutant LRP5 resists sclerostin/DKK1, causing constitutive Wnt/ $\beta$ -catenin signaling [in vitro + mouse]

The gain-of-function is a loss of inhibition. In vitro, six different HBM-LRP5 mutants are resistant: *"neither Dickkopf1 (DKK1) nor sclerostin efficiently inhibits HBM-LRP5 signaling"* (PMID: 18521528); DKK1 and sclerostin act as independent regulators of LRP5, and both are impaired by the HBM mutations. This receptor-resistance model was confirmed in vivo with knock-in mice: *"the (8kb) Dmp1-SOST transgene significantly lowered whole-body bone mineral density (BMD), bone mineral content (BMC), femoral and vertebral trabecular bone volume fraction (BV/TV), and periosteal bone-formation rate (BFR) in wild-type mice but not in mice with Lrp5 p.G171V and p.A214V alleles"* (PMID: 25808845). The p.A214V allele was more sensitive to DKK1 than p.G171V, indicating **mutation-specific, differential inhibitor sensitivity**.

Critically, the osteoclast is a bystander. Osteoclasts from a **T253I** ADO-I patient showed *"normal osteoclastogenesis, expression of osteoclast markers, morphology, and localization of proteins involved in bone resorption, such as ClC-7 and cathepsin K. The ability to resorb bone was also normal"* in vitro, though they showed decreased resorption and low osteoclast numbers in vivo — consistent with an osteoblast-driven defect that secondarily reduces osteoclastogenic support (PMID: 16251418). Circulating levels of the inhibitors themselves are normal — *"There were no significant differences found in the serum levels of sclerostin (SOST), Dickkopf-1 (Dkk-1), or secreted frizzled-related protein-4 (SFRP-4) in affected vs. unaffected individuals"* (PMID: 24927689) — confirming the lesion is at the receptor, not in ligand abundance.

## Finding 3 — Clinical phenotype: generalized osteosclerosis maximal at the cranial vault, often asymptomatic, uniquely NOT fracture-prone [human clinical]

ADO-I *"is characterized by a generalized osteosclerosis, most pronounced at the cranial vault";* patients *"are often asymptomatic but some suffer from pain and hearing loss,"* and — the pathognomonic clinical fact — *"ADOI is the only type of osteopetrosis not associated with an increased fracture rate"* (PMID: 12054167). Across the overlapping LRP5-HBM/Worth-type endosteal-hyperostosis spectrum (155 reported patients), *"Neurological involvement and increased serum alkaline phosphatase (ALP) were present in 19.4% and 3.7% of cases, respectively. Facial changes and torus palatinus were observed in 61% and 41% of cases, respectively"* (PMID: 37659026).

The contrast with **ADO Type II** (Albers-Schönberg disease, caused by *CLCN7*) is diagnostic: there, "*Fractures were common (78% of patients) and healed slowly,*" with sandwich vertebrae and frequent orthopedic surgery (PMID: 10617161). The Wnt-built ADO-I bone is stronger, not more brittle.

#### **Finding 4 — Markedly increased, age-progressive, cortical-predominant BMD; skull-vault thickening is the defining radiographic feature [human clinical]**

Bollerslev and Andersen's original classification of 34 patients showed "*Type I showed a pronounced sclerosis of the skull with an enlarged thickness of the cranial wall*"; radiogrammetry demonstrated normal subperiosteal width but significantly enlarged cortical thickness ( $P < 0.01$ ) with a reduced medullary cavity, and serum acid phosphatase — a resorption marker — was markedly increased in Type II but *not* Type I ( $P < 0.01$ ) (PMID: 3377922). DXA of the entire skeleton quantified the magnitude: "*Median BMD was markedly increased in the axial skeleton by 51% (44-56) and 42% (33-56), (median differences with 95% CI), respectively, for types I and II compared to normal controls,*" with **no overlap between patient and control ranges** and a positive age-BMD correlation in ADO but not controls, indicating progressive osteosclerosis with aging (PMID: 7605703).

#### **Finding 5 — Epidemiology & inheritance: AD, incompletely penetrant, variably expressed; prevalence $\approx 5.5/100,000$ [human clinical]**

The Danish Funen County ascertainment gives the best population estimate: "*the prevalence was 5.5/100,000 inhabitants. The study disclosed 33 patients of whom 32 had the mild, autosomal dominant form of osteopetrosis. Two obligate carriers, who had the genotype but were not phenotypically affected*" — direct evidence of **incomplete penetrance** (PMID: 3829443). Expressivity is wide and age-dependent: "*39% were asymptomatic. The age of first appearance of symptoms also varied widely (8-76 years), with a tendency to increasing symptoms with aging. The frequency of fractures was low*" (PMID: 3829443). Transmission is autosomal dominant through heterozygous *LRP5* gain-of-function variants (PMID: 12579474); each child of an affected parent has a 50% chance of inheriting the allele, though penetrance is not guaranteed.

## Finding 6 — Mouse models: HBM knock-ins reproduce strong bone; *Lrp5*-knockout does the opposite [mouse]

The genetic direction of effect is mirrored in mice. Transgenic mice carrying human *LRP5* **G171V** "show a similar phenotype with greater bone mass and biomechanical performance than wild-type mice," and nano-indentation showed "The intrinsic material property that reflected the bone modulus was greater (48%) in the HBM as compared to the NTG mice" — the material basis of fracture resistance (PMID: 20503060). The **A214V** knock-in is "a high bone mass (HBM) mouse (*Lrp5*(+/p.A214V)) that has increased bone strength from enhanced Wnt signaling," and crossing it into an osteogenesis-imperfecta model raised bone mass and strength (PMID: 27297606). Conversely, *Lrp5*-knockout mice have low femoral-neck cross-sectional area and BV/TV, opposite to *Sost*-knockout (PMID: 31437568). Newer transgenic **A241T** mice (the human A242T equivalent) reproduce the HBM phenotype (PMID: 38909879) and even enhance dental-implant osseointegration (PMID: 41025443).

## Finding 7 — Principal morbidity is cranial-nerve compression from skull-base hyperostosis, not fracture [human clinical]

Because bone accrues maximally at the skull, ADO-I morbidity is neuro-compressive. A 26-year-old woman with ADO-I "presented with intractable, electric shock-like facial pain consistent with" trigeminal neuralgia from neurovascular compression compounded by cranial hyperostosis; the report notes "This condition may lead to cranial nerve compression syndromes, with trigeminal neuralgia (TN) occurring infrequently." She was treated durably by staged suboccipital craniectomy/C1 laminectomy plus microvascular decompression, with 5-year sustained relief (PMID: 41431003). More broadly in adult osteopetrosis, "Bone marrow failure, dental abscess, deafness and visual loss are often underestimated and neglected in relation with lack of awareness and expertise" (PMID: 38593953). Because most patients are asymptomatic, the disorder is frequently found incidentally (PMID: 3829443).

## Finding 8 — Structural basis: mutations map to the E1 $\beta$ -propeller, the sclerostin/DKK1 docking module [structural/computational]

The crystal structure of the LRP6 E1  $\beta$ -propeller revealed it to be a **peptide-recognition module**: "human missense mutations that result in bone overgrowth (bone mineral density, or BMD, mutations) cluster to the E1 propeller domain of LRP5," and "the consensus E1 binding sequence is a close match to a conserved tripeptide motif present in all Wnt inhibitors that bind LRP5/6. We show that this motif is important for DKK1 and SOST binding to LRP6 and for

*inhibitory function, providing a detailed structural explanation for the effect of the BMD mutations" (PMID: 21944579). This is the mechanistic keystone: all ADO-I mutations (G171V, D111Y, G171R, A214T/V, A242T, T253I) sit in the exact surface where the inhibitors' Nxi tripeptide motif must dock, so the mutant receptor cannot be silenced (PMID: 12579474, PMID: 18521528).*

### **Finding 9 — Diagnosis is radiologic + *LRP5* sequencing, with low/normal bone-turnover chemistry [human clinical]**

*"The diagnosis is radiologic, supported by biochemical and genetic examination to identify mutations in the key genes involved in osteoclasts, *TCIRG1*, *CLCN7*, *OSTM1*, *SNX10*, *RANK*, and *RANKL*" — and, for the HBM/Type I form, *LRP5* (PMID: 42096006). Radiographs/CT show generalized osteosclerosis maximal at the cranial vault with cortical thickening and reduced medullary cavity (PMID: 3377922); DXA shows markedly elevated BMD with no overlap with controls (PMID: 7605703). Biochemistry shows normal/low serum acid phosphatase (elevated in Type II), consistent with low turnover (PMID: 3377922, PMID: 3829443). Circulating SOST/DKK1/SFRP-4 are normal and are **not** diagnostic biomarkers (PMID: 24927689). Radiologists play a key role and must anticipate subtype-specific complications (PMID: 38483591).*

### **Finding 10 — Management is supportive/surgical; the mechanism validates anti-sclerostin/anti-DKK1 osteoporosis drugs [human clinical + mouse]**

There is no curative drug. *"Treatment is at present largely the management of complications and includes vitamin D and calcium supplements, IFN- $\gamma$  therapy, and hematopoietic stem cell transplantation (HSCT), the latter being the treatment of choice for most forms of osteopetrosis"* — but HSCT targets the osteoclast-defect (recessive) forms and is mechanistically inapplicable to *LRP5*-driven ADO-I (PMID: 42096006). Symptomatic cranial-nerve compression is managed surgically with durable benefit — *"A staged decompressive approach represents an effective and durable treatment option"* (PMID: 41431003). The ADO-I mechanism is the therapeutic converse of anabolic osteoporosis therapy: *"combined administration of sclerostin and *Dkk1* antibody in WT mice produced a synergistic effect on bone gain"* (PMID: 29875318) — the very biology that ADO-I patients embody constitutively. Prevention is limited to genetic counseling and cascade testing/surveillance (PMID: 3829443).



## Full Section-by-Section Report

### 1. Disease Information

ADO-I is a rare, mild, autosomal-dominant sclerosing bone dysplasia characterized by generalized osteosclerosis most pronounced at the cranial vault, elevated bone mineral density that increases with age, and — uniquely among the osteopetroses — no increased fracture rate. It is now classified as a **high-bone-mass (HBM) disorder** driven by *LRP5* Wnt-co-receptor gain of function rather than as an osteoclast-resorption defect.

| Resource | Identifier                                    |
|----------|-----------------------------------------------|
| OMIM     | #607634 (Osteopetrosis, autosomal dominant 1) |
| Orphanet | ORPHA:53 (Osteopetrosis, autosomal dominant)  |
| MeSH     | D010022 (Osteopetrosis)                       |
| ICD-10   | Q78.2 (Osteopetrosis)                         |
| ICD-11   | FB80.20 (Osteopetrosis)                       |
| Gene     | <i>LRP5</i> (HGNC:6697), 11q13                |

**Synonyms / related terms:** Osteopetrosis autosomal dominant type 1; ADO-I; *LRP5*-related high bone mass; Worth-type autosomal dominant endosteal hyperostosis (overlapping spectrum); "benign osteopetrosis" (a partly misleading label, since cranial complications can be serious).

**Source of information:** Primarily aggregated disease-level resources (OMIM, Orphanet) plus published family and cohort studies; the strongest epidemiologic data derive from a systematic Danish population ascertainment ([PMID: 3829443](#)).

### 2. Etiology

**Causal factor — genetic:** Heterozygous **gain-of-function missense mutations in *LRP5***. Reported alleles include **G171V** (the archetypal HBM mutation), **D111Y**, **G171R**, **A214T**, **A214V**, **A242T**, and **T253I** ([PMID: 12579474](#)). All localize to the N-terminal first  $\beta$ -propeller (E1) domain ([PMID: 21944579](#)).

**Genetic risk factors:** The causal variant itself is the risk factor; a single heterozygous E1-domain allele is sufficient. No independent susceptibility loci are established. Modifier effects on magnitude and skeletal-surface selectivity differ between alleles (e.g., G171V vs A214V), but combining two HBM alleles does not exceed either alone ([PMID: 42113468](#)).

**Environmental risk factors:** None causally established. Age is a strong *modifier of expression* — BMD and symptoms increase with age ([PMID: 7605703](#), [PMID: 3829443](#)).

**Protective factors:** Not applicable in the conventional sense. Notably, the mutation itself is "protective" against fracture and osteoporosis — the same biology exploited pharmacologically by anti-sclerostin drugs.

**Gene–environment interactions:** In *Lrp5*<sup>A214V</sup> mice, the HBM genotype altered the vitamin D endocrine system to maintain high bone mass even under dietary calcium restriction, showing genotype-driven reshaping of calcium handling ([PMID: 38477773](#)). This is model-organism evidence of a GxE interaction on peak bone mass.



### 3. Phenotypes

| Phenotype                                                | Type                   | Onset / Course                        | Frequency                            | Suggested HPO                                                                  |
|----------------------------------------------------------|------------------------|---------------------------------------|--------------------------------------|--------------------------------------------------------------------------------|
| Generalized osteosclerosis, maximal cranial vault        | Radiographic/ physical | Childhood–adult, progressive with age | Defining feature                     | HP:0011001 (Increased bone mineral density); HP:0002684 (Thickened calvaria)   |
| Increased bone mineral density                           | Lab/imaging            | Age-progressive                       | Universal (no overlap with controls) | HP:0011001                                                                     |
| Asymptomatic status                                      | —                      | —                                     | ~39% of carriers                     | —                                                                              |
| Hearing loss / deafness                                  | Clinical sign          | Adult                                 | Subset (cranial-nerve)               | HP:0000365 (Hearing impairment)                                                |
| Cranial-nerve palsy / facial pain (trigeminal neuralgia) | Symptom                | Adult                                 | Uncommon but serious                 | HP:0100259 (Cranial nerve compression); HP:0010829 (Abnormal trigeminal nerve) |
| Visual compromise / optic involvement                    | Clinical sign          | Adult                                 | Uncommon                             | HP:0000505 (Visual impairment)                                                 |
| Facial changes / torus palatinus                         | Physical               | Adult                                 | 61% / 41% (HBM spectrum)             | HP:0001999 (Abnormal facial shape)                                             |
| Bone pain                                                | Symptom                | Variable                              | Subset                               | HP:0002653 (Bone pain)                                                         |
| Low fracture frequency                                   | —                      | —                                     | Low; NOT increased                   | (absence of HP:0002659)                                                        |

**Severity/progression:** Mild, chronic, progressive osteosclerosis; highly variable expressivity with wide age of first symptom (8–76 years) (PMID: 3829443). **Quality-of-life impact:** Often negligible (asymptomatic majority); when cranial-nerve compression occurs, impact can be significant (chronic pain, deafness, visual loss) but is frequently amenable to surgery (PMID: 41431003, PMID: 38593953).

## 4. Genetic / Molecular Information

**Causal gene:** *LRP5* (low-density-lipoprotein-receptor-related protein 5), HGNC:6697, OMIM 603506, chromosome 11q13. *Pathogenic variants:* Heterozygous missense — G171V, D111Y, G171R, A214T, A214V, A242T, T253I — all in the E1  $\beta$ -propeller (PMID: 12579474). *Classification:* pathogenic/likely-pathogenic under ACMG (well-established functional gain-of-function, cosegregation, absence in controls). *Variant type:* missense/single-nucleotide. *Allele frequency:* extremely rare/absent in gnomAD (private/family-specific). *Origin:* germline. *Functional consequence:*\* gain of function via loss of inhibitor (SOST/DKK1) binding at E1 (PMID: 18521528, PMID: 21944579).

The loss-of-function pole of the same gene causes **osteoporosis-pseudoglioma syndrome (OPPG)** — a low-bone-mass phenotype — establishing *LRP5* as a bidirectional bone-mass rheostat (PMID: 40940800). In silico analysis of 17 *LRP5* nsSNPs found the gain-vs-loss outcome is "primarily dictated by the interaction between the molecule and *LRP5*, rather than the specific amino acid substitution" (PMID: 40255261).

**Modifier genes:** allele identity modifies magnitude/surface selectivity (PMID: 42113468). **Epigenetic/chromosomal abnormalities:** none established. **Suggested annotations:** *LRP5* (HGNC:6697); Wnt-activated receptor activity (GO:0042813).

## 5. Environmental Information

No environmental toxin, radiation, lifestyle, or infectious agent causes ADO-I; it is a monogenic disorder. The only environmental interaction documented is dietary calcium: HBM-genotype mice maintain high bone mass under calcium restriction by altering vitamin D metabolism (PMID: 38477773). Infectious agents: not applicable.

## 6. Mechanism / Pathophysiology

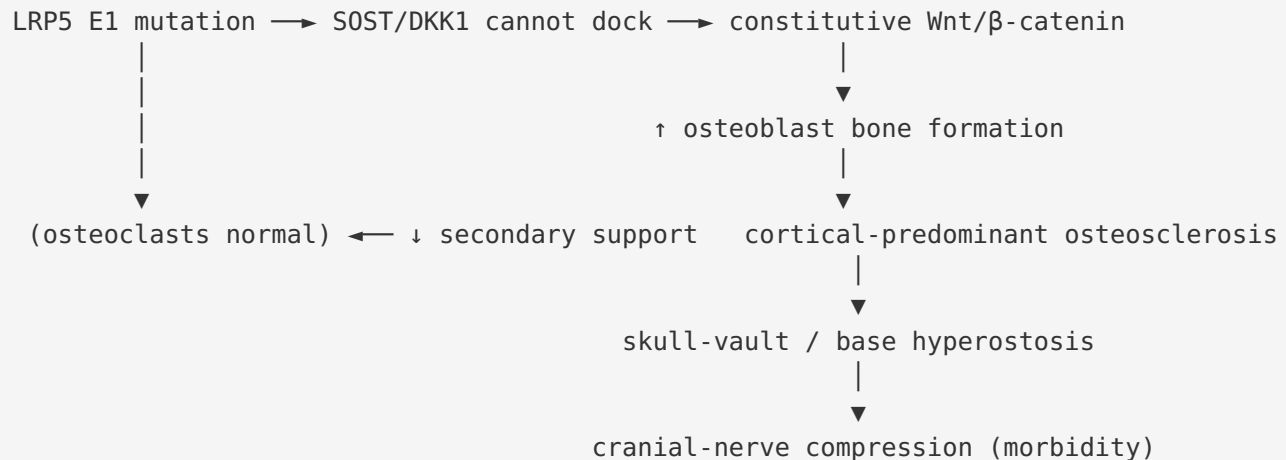
**Ordered causal chain (initiating lesion → clinical manifestation):**

1. A heterozygous missense mutation in the E1  $\beta$ -propeller of *LRP5* (e.g., G171V) alters the surface where Wnt-inhibitor peptides dock (PMID: 12579474, PMID: 21944579).

2. This *prevents sclerostin (SOST) and DKK1* from binding and inhibiting the receptor (PMID: 18521528; confirmed in vivo, PMID: 25808845).
3. Loss of inhibition *results in constitutive Wnt/ $\beta$ -catenin signaling* in osteoblast-lineage cells ( $\beta$ -catenin stabilization  $\rightarrow$  nuclear translocation  $\rightarrow$  RUNX2/Osterix activation) (PMID: 17395698, PMID: 39447984).
4. Sustained signaling *leads to increased osteoblast proliferation, differentiation, and bone-formation rate* (PMID: 25808845).
5. This *produces generalized, age-progressive, cortical-predominant osteosclerosis with increased intrinsic material modulus* — hence strong, fracture-resistant bone (PMID: 3377922, PMID: 7605703, PMID: 20503060).
6. *Branch (osteoclast)*: osteoblast-derived signals secondarily reduce osteoclast support in vivo, though osteoclasts are intrinsically normal (inferred from patient-cell studies) (PMID: 16251418).
7. Bone accrual is **maximal at the skull vault and base**, which *causes narrowing of cranial foramina and posterior-fossa crowding* (PMID: 12054167).
8. This *leads to cranial-nerve compression* — hearing loss, trigeminal neuralgia, facial palsy, visual compromise — the principal source of morbidity (PMID: 41431003, PMID: 38593953).

**Pathway/process detail:** Canonical Wnt/ $\beta$ -catenin (KEGG hsa04310; Reactome Wnt signaling).

**Cellular processes:** osteoblast differentiation (GO:0001649), positive regulation of bone mineralization (GO:0030501), Wnt signaling pathway (GO:0016055), positive regulation of osteoblast differentiation (GO:0045669). **Protein dysfunction:** gain of function via impaired inhibitor docking (not misfolding/aggregation). **Cell types:** osteoblasts (CL:0000062), osteocytes (CL:0000137, the SOST source), osteoclasts (CL:0000092, bystander). **Subcellular:** plasma-membrane receptor (GO:0005886); intriguingly the G171V receptor has poor membrane localization yet still causes HBM, so intracellular retention is unlikely to be the operative mechanism (PMID: 42113468).



## 7. Anatomical Structures Affected

- **Organ/system level:** Skeletal system (primary). Cranial vault and skull base most severely affected (UBERON:0001474 bone element; UBERON:0004339 cranial skeletal system). Secondary: cranial nerves (nervous system) via compression; auditory apparatus (hearing loss); optic pathway (visual compromise); dental structures (torus palatinus).
- **Tissue/cell level:** Cortical bone predominantly (thickened cortex, reduced medullary cavity, [PMID: 3377922](#)); osteoblasts (CL:0000062) are the driver cell; osteocytes (CL:0000137) as inhibitor source; osteoclasts (CL:0000092) intrinsically normal.
- **Subcellular:** LRP5 at the plasma membrane (GO:0005886);  $\beta$ -catenin nuclear signaling (GO:0005634 nucleus).
- **Localization/laterality:** Bilateral, symmetric, generalized osteosclerosis, cranially accentuated.

## 8. Temporal Development

- **Onset:** Variable — childhood through adulthood; the classic AD form often becomes radiographically apparent in adolescence/early adulthood but ranges from 8 to 76 years for first symptoms ([PMID: 3829443](#), [PMID: 38483591](#)). Onset pattern is insidious/chronic.
- **Progression:** Slowly progressive; BMD rises with age (positive age–BMD correlation absent in controls) ([PMID: 7605703](#)). Course is stable-to-progressive, lifelong, not relapsing-remitting. No spontaneous remission.
- **Critical periods:** Peak-bone-mass accrual during growth is amplified by the genotype ([PMID: 38477773](#)); surgical intervention windows arise when cranial-nerve compression becomes symptomatic.

## 9. Inheritance and Population

- **Epidemiology:** Prevalence of the mild AD osteopetrosis form  $\approx 5.5/100,000$  in a Danish population survey (32/33 patients had the AD form) (PMID: 3829443). Incidence not separately quantified.
- **Inheritance:** Autosomal dominant, heterozygous *LRP5* gain of function (PMID: 12579474).
- **Penetrance:** Incomplete — two obligate carriers were phenotypically unaffected (PMID: 3829443).
- **Expressivity:** Highly variable, age-dependent (PMID: 3829443).
- **Anticipation / mosaicism / founder effects:** Not established (non-repeat disorder). Certain alleles (e.g., A242T) recur across multiple families (found in 9 families), but true founder effects are unproven (PMID: 38909879).
- **Consanguinity / carrier frequency:** Not relevant (dominant); "carrier"  $\approx$  affected allele carrier.
- **Demographics:** Reported across populations (notably Danish kindreds for ADO-I); no strong sex bias documented; age distribution skews toward detection in adolescence/adulthood.

## 10. Diagnostics

- **Imaging (primary):** Skeletal radiographs and CT show generalized osteosclerosis maximal at the cranial vault with cortical thickening and reduced medullary cavity; DXA confirms markedly elevated BMD (no overlap with controls) (PMID: 3377922, PMID: 7605703). Distinguish from ADO-II's rugger-jersey/sandwich vertebrae and endobones (PMID: 10617161).
- **Laboratory:** Low bone turnover — serum acid phosphatase normal/low in Type I (elevated in Type II); phosphate occasionally low (PMID: 3377922, PMID: 3829443). Serum SOST/DKK1/SFRP-4 are normal — **not** diagnostic biomarkers (PMID: 24927689).
- **Genetic testing:** Targeted *LRP5* single-gene sequencing (E1 domain) or sclerosing-bone-dysplasia gene panel; WES/WGS where the panel is negative. Radiologic-plus-genetic paradigm (PMID: 42096006). Karyotype/CMA/FISH/mtDNA/repeat testing: not applicable.
- **Differential diagnosis:** ADO-II (*CLCN7*, fracture-prone, sandwich vertebrae); pycnodysostosis; sclerosteosis/van Buchem (*SOST*); Camurati-Engelmann (*TGFB1*); Worth endosteal hyperostosis (*LRP5* spectrum overlap) (PMID: 18328982, PMID: 37659026).
- **Screening:** Cascade genetic testing of first-degree relatives after an index diagnosis.

## 11. Outcome / Prognosis

- **Survival/mortality:** Life expectancy is essentially normal; ADO-I is a "benign" osteopetrosis, in contrast to malignant infantile (recessive) osteopetrosis, though the "benign" label understates cranial risk (PMID: 38483591).
- **Morbidity/function:** Driven by cranial-nerve compression (hearing loss, trigeminal neuralgia, visual loss), dental abscess, and rare marrow issues — "*often underestimated and neglected*" (PMID: 38593953). Fractures are NOT increased and bone is intrinsically strong (PMID: 12054167, PMID: 20503060).
- **Recovery:** Surgical decompression of cranial-nerve compression can give durable relief (5-year sustained) (PMID: 41431003).
- **Prognostic factors:** Location/severity of cranial hyperostosis and specific allele (magnitude/surface selectivity differ) (PMID: 42113468). No validated molecular prognostic biomarker.

## 12. Treatment

- **Pharmacotherapy:** No curative or disease-modifying drug. Supportive vitamin D/calcium; complication management (PMID: 42096006). Anti-resorptives (bisphosphonates/denosumab) are not indicated — resorption is not the problem. (NCIT: vitamin D therapy, calcium supplement.)
- **HSCT:** The treatment of choice for osteoclast-defect (recessive) osteopetroses, but **mechanistically inapplicable** to LRP5/osteoblast-driven ADO-I (PMID: 42096006). (NCIT: hematopoietic stem cell transplantation.)
- **Surgical/interventional:** Decompressive surgery for cranial-nerve compression (e.g., staged posterior-fossa decompression + microvascular decompression for trigeminal neuralgia) — durable (PMID: 41431003); orthopedic surgery for the rare fracture is technically demanding in sclerotic bone. (NCIT: surgical procedure; microvascular decompression.)
- **Supportive/rehabilitative:** Audiology, ophthalmology, dental care, pain management, multidisciplinary follow-up (PMID: 38593953).
- **Therapeutic significance (mirror image):** ADO-I biology validated the **anti-sclerostin (romosozumab)** and **anti-DKK1** anabolic osteoporosis strategy — blocking these inhibitors phenocopies the ADO-I gain of function and synergistically raises bone mass (PMID: 29875318, PMID: 17395698, PMID: 39447984).

### 13. Prevention

- **Primary prevention:** Not possible (monogenic, germline). **Genetic counseling** is the mainstay — autosomal dominant, 50% transmission risk per child, with the caveat of incomplete penetrance and variable expressivity ([PMID: 3829443](#)).
- **Secondary prevention:** Cascade testing and periodic surveillance (audiology, ophthalmology, neuro-imaging) to catch cranial-nerve compression early.
- **Tertiary prevention:** Timely surgical decompression to prevent permanent neural deficit ([PMID: 41431003](#)).
- **Reproductive options:** Preimplantation/prenatal genetic testing available where a familial variant is known. Immunization/public-health/environmental measures: not applicable.

### 14. Other Species / Natural Disease

- **Taxonomy/orthologs:** *LRP5* is conserved; mouse *Lrp5* (NCBI Gene 16973; MGI:1278315) is the principal ortholog used to model the disease.
- **Natural disease:** No well-characterized spontaneous ADO-I analog is documented in companion animals in the reviewed literature; the strongest cross-species evidence is engineered mouse models (below). Comparative biology: the Wnt/LRP5 bone-mass axis is evolutionarily conserved, and mouse HBM alleles faithfully reproduce the human direction of effect.
- **Zoonotic potential:** Not applicable (non-infectious, genetic).



## 15. Model Organisms

| Model                                                 | Type                   | Phenotype recapitulation                                                   | Reference                                                          |
|-------------------------------------------------------|------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------|
| Human <i>LRP5</i> G171V transgenic mouse              | Transgenic (mammalian) | HBM, greater bone mass & biomechanical performance; +48% intrinsic modulus | <a href="#">PMID: 20503060</a>                                     |
| <i>Lrp5</i> p.G171V & p.A214V knock-in mice           | Knock-in               | HBM; resistant to SOST/DKK1-induced osteopenia                             | <a href="#">PMID: 25808845</a>                                     |
| <i>Lrp5</i> p.A214V knock-in                          | Knock-in               | Increased bone mass & strength via enhanced Wnt; rescues OI model          | <a href="#">PMID: 27297606</a>                                     |
| <i>Lrp5</i> A241T transgenic ( $\approx$ human A242T) | Transgenic             | HBM phenotype reproduced; enhances implant osseointegration                | <a href="#">PMID: 38909879</a> ,<br><a href="#">PMID: 41025443</a> |
| <i>Lrp5</i> knockout                                  | Knockout (contrast)    | LOW bone mass (opposite pole) — confirms directionality                    | <a href="#">PMID: 31437568</a>                                     |
| Combined G171V/A214V knock-in                         | Knock-in               | No additive benefit; probes surface-selective mechanism                    | <a href="#">PMID: 42113468</a>                                     |

**Applications:** Dissecting Wnt-inhibitor resistance, bone material properties, GxE (dietary calcium/vitamin D), and validating anti-SOST/anti-DKK1 anabolic strategies. **Limitations:** mouse models capture the HBM/strength phenotype well but do not fully model human skull-base cranial-nerve compression; the G171V membrane-localization paradox remains unresolved ([PMID: 42113468](#)). **Resources:** MGI (mouse *Lrp5*, MGI:1278315).

## Mechanistic Model / Interpretation

ADO-I is best understood as a **single-node dysregulation of the Wnt/LRP5 bone-mass rheostat**. LRP5 integrates activating Wnt ligands against inhibitory SOST and DKK1 at its E1  $\beta$ -propeller. In health, osteocyte-derived sclerostin tunes down osteoblast Wnt signaling to keep bone formation in check. ADO-I mutations physically remove the inhibitor "off-switch" from a single receptor surface, so the osteoblast receives an unopposed "build" signal for life. The

result is more bone — and, because Wnt-built bone has superior intrinsic material properties, *stronger* bone. This is the mechanistic explanation for the single most distinctive clinical feature of the disease: **it is the only osteopetrosis without excess fractures.**

The same node explains the disease's paradoxes: - **Osteoclasts are normal** because the lesion is upstream in the osteoblast; osteoclast changes seen in vivo are secondary. - **Skull-predominant morbidity** follows because the skull is where excess formation causes the most mechanical mischief (foraminal narrowing), not because the skull is uniquely susceptible at the molecular level. - **Bidirectionality of *LRP5*** (OPPG at the loss-of-function pole, HBM at the gain-of-function pole) makes ADO-I a natural human experiment that de-risked anti-sclerostin/anti-DKK1 drug development — patients constitutively embody the very pharmacology those drugs deliver.

| Axis           | ADO-I (gain of function)   | ADO-II / <i>CLCN7</i> (contrast) | <i>LRP5</i> -LOF / OPPG (opposite pole) |
|----------------|----------------------------|----------------------------------|-----------------------------------------|
| Defective cell | Osteoblast (Wnt signaling) | Osteoclast (acidification)       | Osteoblast (Wnt signaling)              |
| Bone mass      | ↑ ↑ (age-progressive)      | ↑ ↑                              | ↓ ↓                                     |
| Fracture risk  | <b>Not increased</b>       | Increased (78%)                  | Increased                               |
| Skull vault    | Marked thickening          | Variable                         | Low mass                                |
| HSCT           | Not applicable             | Not curative in AD form          | Not applicable                          |

## Evidence Base

| PMID                     | Contribution                                                               | Supports        |
|--------------------------|----------------------------------------------------------------------------|-----------------|
| <a href="#">12054167</a> | Maps ADO-I to 11q12-13; defines cranial-vault phenotype & no-fracture rule | F001, F003      |
| <a href="#">12579474</a> | Six novel N-terminal <i>LRP5</i> HBM mutations                             | F001, F008      |
| <a href="#">20855225</a> | Reclassification of ADO-I as HBM disease                                   | F001            |
| <a href="#">18521528</a> | HBM-LRP5 resists DKK1/sclerostin (in vitro)                                | F002            |
| <a href="#">25808845</a> | In vivo proof of inhibitor resistance in knock-ins                         | F002, F006      |
| <a href="#">16251418</a> | T253I osteoclasts intrinsically normal                                     | F002            |
| <a href="#">24927689</a> | Circulating SOST/DKK1/SFRP4 normal                                         | F002, F009      |
| <a href="#">21944579</a> | E1 propeller = inhibitor docking; mutations cluster there                  | F008            |
| <a href="#">3377922</a>  | Original two-type classification; cortical thickening                      | F004, F009      |
| <a href="#">7605703</a>  | DXA: +51% axial BMD, age-progressive                                       | F004            |
| <a href="#">3829443</a>  | Prevalence 5.5/100k, incomplete penetrance, 39% asymptomatic               | F005            |
| <a href="#">20503060</a> | G171V transgenic: +48% modulus, superior strength                          | F006            |
| <a href="#">27297606</a> | A214V knock-in improves strength; rescues OI                               | F006, F010      |
| <a href="#">10617161</a> | ADO-II fracture-prone contrast (78%)                                       | F003            |
| <a href="#">37659026</a> | LRP5-HBM spectrum phenotype frequencies                                    | F003, F007      |
| <a href="#">41431003</a> | Trigeminal neuralgia; durable surgical decompression                       | F007, F010      |
| <a href="#">38593953</a> | Underrecognized adult osteopetrosis complications                          | F007            |
| <a href="#">42096006</a> | Diagnostic/therapy paradigm; HSCT scope                                    | F009, F010      |
| <a href="#">29875318</a> | Anti-SOST + anti-DKK1 synergy (mirror image)                               | F010            |
| <a href="#">42113468</a> | Allele-specific magnitude/surface effects; G171V paradox                   | F006, mechanism |

Supporting/context papers: [17395698](#), [39447984](#), [40940800](#), [40255261](#), [38909879](#), [41025443](#), [38477773](#), [31437568](#), [18328982](#), [38483591](#).

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## Limitations and Knowledge Gaps

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1. **Nosological overlap.** ADO-I, LRP5-HBM, and Worth-type endosteal hyperostosis form a continuum; some cited frequencies (e.g., facial changes 61%, torus palatinus 41%) derive from the broader spectrum rather than genetically confirmed ADO-I alone ([PMID: 37659026](#)).
  2. **Epidemiology is dated and geographically narrow.** The 5.5/100,000 prevalence and penetrance estimates rest chiefly on one Danish survey ([PMID: 3829443](#)); modern multi-ethnic, genotype-confirmed prevalence and sex-ratio data are lacking.
  3. **Genotype–phenotype correlation is incomplete.** Alleles differ in magnitude and skeletal-surface selectivity, but predictors of who develops cranial-nerve compression are unknown ([PMID: 42113468](#)).
  4. **Mechanistic loose end.** The G171V receptor causes HBM despite poor plasma-membrane localization — its mechanism "*remains elusive*" ([PMID: 42113468](#)).
  5. **No disease-specific therapy or trials.** Management is generic/supportive; no ADO-I-targeted pharmacology (e.g., a rational Wnt-dampening approach) has been tested.
  6. **Biomarkers absent.** Circulating Wnt inhibitors are not diagnostic or prognostic ([PMID: 24927689](#)); no molecular marker predicts cranial complications.
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## Proposed Follow-up Experiments / Actions

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1. **Genotype-stratified natural-history registry.** Prospectively enroll genetically confirmed ADO-I patients to quantify per-allele penetrance, age-of-onset distributions, sex ratio, and the incidence/timing of cranial-nerve compression.
2. **Predictive imaging biomarkers.** Correlate quantitative skull-base CT foraminal metrics with cranial-nerve outcomes to build a risk-stratification tool that triggers earlier prophylactic decompression.

3. **Resolve the G171V paradox.** Use surface-biotinylation, live-cell receptor trafficking, and structural studies to determine how a poorly membrane-localized receptor drives HBM (PMID: 42113468).
4. **Rational therapeutic test in models.** Since ADO-I is Wnt-over-signaling, test whether pharmacologic Wnt-inhibitor delivery (recombinant sclerostin/DKK1 agonism or LRP5-E1-targeting antibodies) can selectively dampen skull hyperostosis in A241T/A214V mice without inducing systemic osteopenia.
5. **Auditory/ophthalmologic surveillance protocol.** Formalize and validate a multidisciplinary screening schedule to catch reversible cranial-nerve deficits early (PMID: 38593953).
6. **Comparative-genetics screen.** Search veterinary/OMIA resources and companion-animal cohorts for naturally occurring LRP5-HBM analogs to broaden the comparative model base.
7. **MONDO/ontology curation.** Formalize the ADO-I MONDO entry and precise HP/UBERON/CL annotations (E1-domain gain of function; cranial-nerve compression; osteoblast-driven cortical osteosclerosis) for knowledge-base integration.

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*Report compiled from a five-iteration autonomous investigation: 10 confirmed findings, 44 papers reviewed. Evidence source types span human clinical (family/cohort/case studies), model organism (knock-in/transgenic/knockout mice), in vitro (receptor-binding assays), and computational (in silico variant analysis, structural biology).*

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